# Xu Han

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Photonics M.Sc. student and LMU astrophysics graduate focused on astronomical instrumentation, wavefront sensing and optical-system performance, combining optical modelling, experimental apparatus development and astronomical data analysis.

## Education

**M.Sc. Photonics** | Munich University of Applied Sciences *Mar 2026 - Present*
**Selected coursework:** Physical Optics and Modelling (**Zemax optical design**); Micro and Fiber Optics; Optical Measurement Technology; Semiconductor Optics and Components.

**M.Sc. Astrophysics** | LMU Munich *Apr 2022 - Aug 2024*
Grade: 1.55 (German scale). **Selected coursework:** Essentials of Advanced Astrophysics; Cosmology and Structure Formation; Statistical Methods; Instrumental Laboratory Courses.

**B.Sc. Physics** | University of Duisburg-Essen *Oct 2018 - Mar 2022*
Grade: 1.8 (German scale).

**B.Eng. Electrical Engineering and Automation** *Sep 2013 - Jul 2017*
Hubei University of Technology

## Selected Research and Engineering Projects

### Shack-Hartmann Wavefront Sensing & Adaptive Optics Simulation

*Independent project (ongoing development)* *Mar 2026 - Present*
Interactive explorer | Source code

- Developed a modular Python SCAO framework connecting atmospheric turbulence, detector-level Shack-Hartmann sensing, calibrated reconstruction, deformable-mirror correction and delayed feedback.
- Investigated detector-noise and gain/latency trade-offs using residual wavefront error, propagated-PSF Strehl and encircled energy.
- Diagnosed a reconstruction failure associated with measurement dropout in a 2.2 m NGS model; stronger TSVD regularisation kept reconstruction usable throughout the matched two-second run.
- Cross-checked selected optical components against HCIPy and analytic references; documented reproducible experiments, model limitations and an interactive explorer of recorded results.

### Experimental Collisional Charging Apparatus

*Mar 2021 - Aug 2021*
*Bachelor's thesis | University of Duisburg-Essen*
Supervisors: Prof. Gerhard Wurm and Dr. Felix Jungmann.

- Designed the mechanical apparatus in CAD, fabricated components by 3D printing and assembled the setup; integrated and debugged electronic charge-measurement modules.
- Conducted measurements and analysed grain-size dependence of collisional charging, testing a dust-collision growth hypothesis relevant to early planet formation.

### Weak Lensing Void Finder

*Mar 2023 - Aug 2024*
*Master's thesis | LMU Munich*
Supervisors: Prof. Joseph Mohr and Dr. Chris Davies.

- Developed and parameter-tested a watershed finder for cosmic voids in 2D full-sky weak-lensing convergence maps.
- Applied the analysis to DES Y3 weak-lensing maps; characterised void abundance, effective radii, ellipticities and convergence profiles.

## Laboratory and Instrumentation Training

- **Photonics laboratory:** Completed experiments: APD measurements, laser spectra, nonlinear optics, laser interferometry, fiber Bragg gratings and fiber coupling.
- **Astronomical instrumentation laboratory:** Completed experiments: CCD imaging and data reduction, optical interferometry, deformable-mirror AO tuning and radio measurements of Galactic rotation.
- **Observatory experience:** Participated in observing and image processing at Wendelstein Observatory using the 43 cm telescope.

## Additional Independent Projects

### Astrophotonics

- Modelled how AO wavefront error, injection jitter and photonic-lantern mode count affect fiber coupling, spectral SNR and radial-velocity precision.

### ELT and OWL Telescope Optics

*Personal project extending a hands-on Zemax lab in Physical Optics and Modeling*

- Reconstructed the two telescope designs in Zemax and compared how aperture, off-axis aberrations and focal-surface shape affect the concentration of starlight.

## Work Experience and Career Break

### Python Developer | BaryonBib — *Dec 2024 - Feb 2025*

*MAX!mize start-up team, MPA, Garching*

- Contributed to a centralised bibliography platform in Python; implemented and tested LLM-based functionality.

**Medical leave** — *Jan 2025 - Dec 2025*

Recovery and rehabilitation following an accident and surgery.

## Talks, Workshops and Schools

**MAVIS 2026 workshop | Participant** — *Sep 2026*

**Tonale Winter School on Cosmology | Participant** — *Dec 2024*

**ORIGINS Lensing Day** — *Nov 2024*

Contributed talk: Watershed on the curved sky

**Cambridge-LMU Cosmology Meeting | Participant** — *Dec 2023*

**Advanced Astrophysical Seminar | LMU Munich** — *Nov 2022*

40-minute course presentation and literature study on large ground-based astronomical telescopes, telescope designs and adaptive optics.

## Technical Skills

**Optical modelling:** Wavefront sensing, Fourier PSFs, calibration, regularised reconstruction and closed-loop diagnostics; HCIPy and TIPTOP; Zemax through optical-design coursework.

**Scientific computing:** Python (NumPy, SciPy, Matplotlib, pandas, scikit-learn); MATLAB; Mathematica; statistical analysis and MCMC.

**Experimental methods:** CAD, 3D printing, mechanical assembly, electronic measurement integration, optical alignment and interferometry.

**Research tools:** Linux, Git, Jupyter and LaTeX.

## Languages and Interests

**Languages:** Chinese (native); English (fluent); German (fluent); French (beginner).
**Interests:** Astronomical photography and alpine mountaineering.

## Academic Reference

Prof. Joseph Mohr, M.Sc. thesis supervisor